



Confidence-Aware Brain Tumor Classification

A Reliability-First Deep Learning System for Medical AI

ResNet18

Temperature Scaling

Grad-CAM

ECE Calibration

Abstention

Repository: github.com/taalch18/neuro_confidence_aware

Document: Full Technical Deep-Dive & Interview Prep Guide

Python 3.10 | PyTorch 2.x | MIT License | 2026

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1. The Problem — Why 'Confident But Wrong' Kills in Medicine

Imagine a radiologist's assistant that looks at an MRI and declares: '**This is definitely a glioma — 94% confidence.**' The doctor, trusting the system, proceeds with an aggressive treatment plan. But the model was wrong. The tumour was a meningioma — far less aggressive, with a completely different treatment protocol. The patient is harmed. This is not hypothetical — it is the central failure mode of uncalibrated neural networks.

The core insight: In medical AI, a wrong prediction with LOW confidence is manageable — it gets flagged, reviewed, escalated. A wrong prediction with HIGH confidence is catastrophic — it bypasses human review. Reducing 'confident errors' is more important than maximising raw accuracy.

The Mathematical Root

Standard networks end with **softmax**, which converts logits into a probability distribution summing to 1. This looks like probability — but it is not calibrated. Cross-entropy loss rewards high confidence on correct predictions, incentivising the model to push logits far apart — even when it should be uncertain. The result: a model that is confidently wrong on edge cases.

Interview Q: Why doesn't softmax give you reliable probabilities?

Softmax is trained with cross-entropy loss, which rewards high confidence on correct predictions. This pushes logits further apart than warranted. The predicted probability reflects logit geometry, not real-world frequency. A calibrated model would have 80% accuracy on cases where it says 80% confident.

2. Project Overview & Aim

This project solves three problems simultaneously: classify brain tumours accurately, know when it does not know, and explain decisions to clinicians. These are the three pillars of trustworthy medical AI.

Pillar	What It Means	How This Project Achieves It
Accuracy	Correct tumour classification	ResNet18 fine-tuned on MRI data (92.3%)
Calibration	Confidence scores reflect real accuracy	Temperature Scaling (ECE: 0.124→0.031)
Safety	Refuse to answer when uncertain	Threshold abstention at confidence < 0.8
Interpretability	Explain which pixels drove the decision	Grad-CAM visualisation

3. System Architecture & Workflow Diagram

The diagram below shows the complete inference pipeline — from a raw MRI scan through preprocessing, the ResNet18 backbone, temperature scaling calibration, the safety abstention gate, and final outputs. Each component is colour-coded by phase. The dashed lines represent the Grad-CAM interpretability branch running in parallel to the main prediction path.

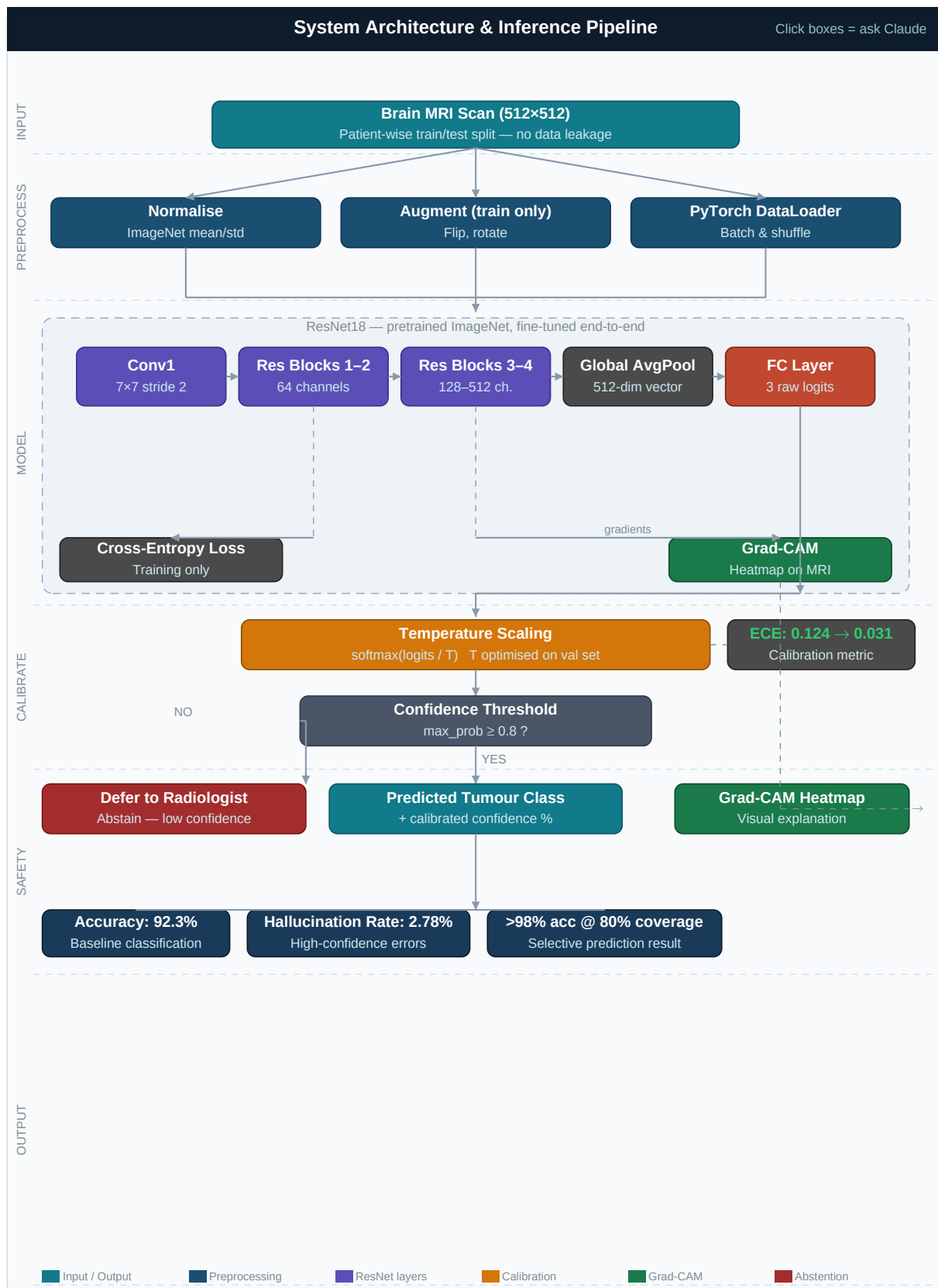


Figure 1: Full system pipeline. Solid arrows = primary data flow. Dashed arrows = auxiliary branches (Grad-CAM gradients, calibration metrics). The abstention gate splits into YES (prediction) and NO (defer to radiologist) paths.

4. The Dataset — What Data, Why It Matters

The project uses the **Cheng et al. Brain Tumor MRI Dataset** (Figshare 1512427), a publicly available benchmark with approximately 3,000 MRI slices across three tumour classes.

Class	Description	Clinical Significance
Meningioma	Arises from meninges (brain lining)	Usually benign; slow-growing
Glioma	Arises from glial cells	Often malignant; aggressive treatment needed
Pituitary Tumour	Arises from pituitary gland	Affects hormonal regulation

The Critical Design Decision: Patient-Wise Split

Most beginners split data randomly — 80% train, 20% test. In medical imaging this causes data leakage: a single patient contributes dozens of slices, so the model learns patient-specific features rather than tumour features. Patient-wise split ensures ALL slices from a given patient go exclusively to either train OR test — forcing generalisation to new patients.

Patient-wise split is the single decision that separates research-quality work from tutorial-quality work. It ensures the model generalises to new patients — the actual clinical deployment scenario.

Interview Q: Why is random train/test splitting wrong for medical data?

Multiple slices from one patient are not independent samples — they share anatomy, noise, and scanner characteristics. Random splitting causes data leakage: the model memorises patient-specific features. Patient-wise splitting forces generalisation to unseen patients — the real deployment condition.

5. Model Architecture — ResNet18 Deep Dive

Why ResNet18 Specifically?

ResNet18 is a Residual Network with 18 layers. Before ResNets (2015), deep networks suffered from vanishing gradients: backpropagated error signals shrink exponentially through many layers. ResNets introduced **skip connections**: instead of learning $F(x)$, each block learns $F(x)+x$. Gradients flow directly through the skip path, bypassing layer stacks.

Architecture	Parameters	Fit for ~3K images?	Verdict
Custom CNN	~0.5M	Possible but limited	Underpowered
ResNet18	11M	Yes — transfer learning helps	OPTIMAL
ResNet50	25M	Risk of overfitting	Overkill
Vision Transformer	86M+	No — needs 100K+ images	Wrong tool
EfficientNet-B4	19M	Borderline	Acceptable alternative

Transfer Learning & Final Layer Modification

ResNet18 is loaded with ImageNet weights. Early layers have learned universal visual features (edges, textures, blobs) directly useful for medical images. The final FC layer is replaced: from 1000 ImageNet classes to 3 tumour classes. Only this layer trains from scratch.

```
model = torchvision.models.resnet18(pretrained=True)
model.fc = nn.Linear(512, 3) # Replace final layer
```

Interview Q: Why not train a CNN from scratch on your 3K dataset?

With ~3,000 images, a CNN from scratch would overfit severely — memorising training images rather than learning general tumour features. Transfer learning from ImageNet gives a head start: early layers already detect edges and textures. We only teach tumour-specific patterns on top.

6. The Calibration Crisis — Softmax's Dirty Secret

What Is Calibration?

A model is **perfectly calibrated** if, among all predictions made with X% confidence, exactly X% are correct. A weather forecaster saying '70% chance of rain' on 100 days is well-calibrated if it rained on roughly 70 of those days.

What Is ECE (Expected Calibration Error)?

1. Collect all predictions on the test set.
2. Bin by confidence — e.g. [0.7–0.8], [0.8–0.9], [0.9–1.0].
3. For each bin: |avg confidence – accuracy in bin|.
4. ECE = weighted average of gaps across bins.

ECE = 0.124 before scaling: on average, the model's confidence exceeds its accuracy by 12.4 percentage points. ECE = 0.031 after scaling: the gap is only 3.1 points — excellent.

If the model says '90% confident' on 100 cases, but is only right 77 times — ECE captures that 13-point gap. Before calibration, this causes dangerous high-confidence errors. After temperature scaling, confidence and accuracy are aligned.

7. Temperature Scaling — The Fix, From Math to Code

Temperature Scaling is post-hoc calibration — applied after training, model weights frozen. It introduces a single scalar T applied to logits before softmax:

Calibrated probability = softmax(logits / T) T is optimised to minimise NLL on the validation set ONLY. Weights never change. One number changes.

When $T > 1$, logits shrink → softmax becomes softer → confidence is more honest.

Raw logits: [4.2, 1.1, 0.3]

Softmax ($T=1.0$): [0.94, 0.04, 0.02] ← overconfident

Scaled logits ($T=1.5$): [2.8, 0.73, 0.2]

Softmax ($T=1.5$): [0.83, 0.11, 0.06] ← calibrated

Method	Parameters	Verdict vs Temp Scaling
Temperature Scaling	1 scalar	Best: simple, effective, post-hoc
Platt Scaling	2	Similar, less principled
Isotonic Regression	Many	Overfits on small validation sets
Monte Carlo Dropout	Existing dropout	Good but requires multiple forward passes
Deep Ensembles	$N \times$ all weights	Best accuracy, $N \times$ compute cost

Interview Q: Temperature Scaling changes nothing in the model weights. How does it improve anything?

The model's logit ordering is already correct — ResNet18 can distinguish tumours. The problem is only in logit magnitude, not rank. Dividing by $T > 1$ is monotonic: argmax is preserved, accuracy stays identical, but confidence distributions become honest. One parameter, fit in seconds.

8. Safety Mechanism — Abstention & Risk-Coverage

Once calibrated, the system implements **selective prediction**: if max class probability after temperature scaling is below 0.8, the model refuses to give a final answer and defers to a human radiologist.

The Risk-Coverage Trade-off

- High threshold (0.95): very low error rate, but many cases deferred.
- Low threshold (0.5): model answers almost everything → higher error rate.
- Threshold = 0.8: model answers ~80% of cases, achieving >98% accuracy on those.

The hallucination rate of 2.78% means only ~3% of the model's errors are high-confidence errors — the model has learned to be uncertain when it is wrong.

Interview Q: How did you choose the threshold of 0.8?

Analyse the risk-coverage curve on a held-out validation set. Plot accuracy vs coverage for all possible thresholds. Find the knee point — where you start deferring confident correct predictions for marginal safety gain. 0.8 gives >98% accuracy on 80% of cases — a clinically reasonable operating point.

9. Interpretability — Grad-CAM Explained

Grad-CAM answers: *which pixels most influenced this prediction?* A radiologist will not act on a model's prediction without understanding what the model looked at.

The Algorithm Step by Step

1. Forward pass → feature maps from last conv layer.
2. Backpropagate target class score → gradients at last conv layer.
3. Global average pool gradients → importance weight α_k per channel.
4. Weighted sum of feature maps $\times \alpha_k$. Apply ReLU.
5. Upsample heatmap (e.g. 14×14) to 512×512 . Overlay on MRI.

No model retraining required. If Grad-CAM highlights tumour regions, the model learned correctly. If it highlights skull edges or background, the model learned spurious correlations — a critical sanity check for any medical AI system.

10. Results & What They Actually Mean

Metric	Value	What It Means	Good Benchmark
Baseline Accuracy	92.3%	Standard classification on test set	>90% solid for 3-class MRI
ECE (Before)	0.124	Overconfident by 12.4pp avg	Needs fixing
ECE (After)	0.031	Calibrated to ~3pp	Excellent (<0.05 is good)
Hallucination Rate	2.78%	Fraction of errors that are high-confidence	Below 5% is the target
Coverage @ 0.8 thresh	~80%	Fraction the model answers	Tunable by threshold
Accuracy on answered	>98%	Accuracy when confident	Selective prediction working

11. Technology Choices & Why Alternatives Lost

Technology	Role	Why Chosen	Best Alternative	Why Alt. Lost
PyTorch	Deep learning framework	Dynamic graphs, research standard	TensorFlow/Keras	Less flexible for custom loops; hides too much
ResNet18	Feature extractor	11M params, pretrained, proven on ImageNet	EfficientNet-S	Similar performance, less transparent pedagogically
Temperature Scaling	Calibration	1 parameter, post-hoc, preserves MC	Monte Carlo Dropout	Needs arch changes; slower inference
Grad-CAM	Interpretability	No model changes, works on any CNN	CIME / SHAP	Slow; SHAP for images is approximate
Threshold Abstention	Safety	Simple, interpretable, maps to clinician workflow	Confidence Prediction	Stronger guarantees but more complex

12. Competitive Analysis — Top 5% Benchmark

The project concept is top-5%. The execution gap is ownership depth — the ability to defend every design decision without hesitation.

Criterion	Bottom 50%	Top 15%	Top 5%	This Project
Domain	MNIST, CIFAR	Standard CV	Medical / scientific	Strong
Technical Depth	Tutorial code	One technique	Multiple integrated	Strong
Data Rigor	Random split	Stratified	Patient-wise, leakage-free	Strong
Evaluation	Accuracy only	F1 + confusion	ECE + selective + hallucination	Strong
Ownership	Cannot explain	High-level only	Defend every decision	GAP — needs work

VERDICT: The **PROJECT CONCEPT** is top-5%. The **EXECUTION GAP** is ownership depth. Close this gap by internalising every section in this document until you can explain it from scratch without notes.

13. Interview Prep — Questions You Will Be Asked

Tier 1: Google / Meta / DeepMind Research Internship

Interview Q: Prove that Temperature Scaling does not change accuracy.

Dividing logits by $T > 0$ is monotonic — it preserves logit ranking. $\text{argmax}(\text{logits}/T) = \text{argmax}(\text{logits})$ for any positive T . Predictions don't change, accuracy doesn't change. Only confidence magnitudes change.

Interview Q: What is the relationship between ECE and the Brier Score?

Both measure calibration but differently. ECE is a bucketed estimator — bins predictions and measures the gap between confidence and accuracy. Brier Score is a proper scoring rule: MSE between predicted probability and true label. Brier Score penalises both miscalibration and inaccuracy simultaneously. ECE is more interpretable; Brier Score more mathematically principled.

Interview Q: Why might Grad-CAM be misleading on ResNet18?

Grad-CAM uses gradients at the last conv layer, which has 14×14 spatial resolution after 5 downsampling stages from 512×512 . Upsampling from 14×14 to 512×512 introduces spatial imprecision. Also, ResNet's skip connections mean gradients flow through multiple paths — Grad-CAM captures only one path.

Tier 2: AI Startups / Applied ML Roles

Interview Q: Walk me through the full inference pipeline from a new MRI to a clinical output.

MRI $\rightarrow$ resize $512 \times 512 \rightarrow$ normalise with ImageNet stats $\rightarrow$ ResNet18 forward pass $\rightarrow$ raw logits $\rightarrow$ divide by learned $T \rightarrow$ softmax $\rightarrow$ calibrated probabilities $\rightarrow$ if max prob < 0.8 : abstain (DEFER TO RADIOLOGIST) $\rightarrow$ else: predicted class + confidence + Grad-CAM heatmap.

Interview Q: Your ECE is 0.031. Is that good enough for clinical deployment?

For research: yes, excellent. For clinical deployment: no. Requires external validation on data from different hospitals and scanners, prospective trials, and regulatory approval (FDA 510k / CE Mark). This is an MVP proof-of-concept, not a product.

Tier 3: Indian Product Companies (ML roles)

Interview Q: What is overfitting and how did you prevent it?

Overfitting = memorising training data rather than learning generalisable features. Prevention: (1) Transfer learning from ImageNet. (2) Data augmentation — flips/rotations. (3) Patient-wise split — no leakage. (4) Early stopping when val loss stops improving.

Interview Q: What would you do if you had 10x more data?

With 30,000 images: (1) Experiment with ResNet50 or EfficientNet-B4. (2) 3D volumetric models — process full MRI volumes rather than 2D slices. (3) Deep Ensembles — train 5 independent models, average predictions. (4) Out-of-distribution detection on healthy brain scans.

14. Weaknesses & How to Address Them

Knowing your project's limitations and having a plan for them separates a researcher from a student. Never hide weaknesses — acknowledge them with conviction.

Weakness	Honest Impact	What You Say in Interview
Small dataset (~3K)	Limits generalisability; high variance	Acknowledged. Patient-wise split mitigates. Future: multi-institution data.
2D slice-by-slice	Misses 3D volumetric context	Known limitation. 3D CNN is next step. 2D is standard MVP approach.
Internal validation only	No external clinical validation	MVP only. Clinical deployment requires multi-site validation + regulatory approval.
No class imbalance handling	Bias toward frequent classes	Should add weighted loss or oversampling. Known gap for v2.
Heuristic threshold	0.8 not statistically principled	Conformal Prediction gives distribution-free guarantees. Strong future work.

15. Suggested Resume Descriptions

For ML/AI Research Internships (Google Brain, DeepMind, etc.)

Confidence-Aware Brain Tumor Classification | PyTorch, ResNet18, Python Built a reliability-first medical AI system for 3-class MRI tumour classification. Implemented post-hoc Temperature Scaling to reduce Expected Calibration Error from 0.124 to 0.031, achieving <3% high-confidence error rate. Designed patient-wise data splitting to prevent leakage. Integrated Grad-CAM for interpretability and threshold-based selective prediction achieving >98% accuracy at 80% coverage.

For Software Engineering / General Tech Internships

Medical Image Classification System | PyTorch, Python, ResNet18 Developed a brain tumour classifier on MRI data (92.3% accuracy, 3 classes). Applied Temperature Scaling calibration to improve confidence reliability. Built a safety-first inference pipeline with automatic deferral for uncertain cases, reducing dangerous high-confidence errors to <3%. Added Grad-CAM visualisation for model explainability.

One-Liner (for skills/project list)

Brain Tumor MRI Classifier with calibrated confidence scoring (ECE: 0.031) and selective prediction — PyTorch, ResNet18, Temperature Scaling, Grad-CAM

FINAL NOTE: The gap between top-15% and top-5% is not the project — it's you. Study this document until you can explain every section without notes. Then explain it to someone who has never heard of neural networks. If you can do both, you are ready.